The Galton board: findings

West bay 3: the Galton board · 14 November 2026

How the board came to be as it is, and what was measured. MACHINES.md had it as "balls and pins only; easy". The pins were easy enough; getting the balls in was not. Unless said, every run was at 2400 steps a second and 20 solver passes, 150 balls of 6 mm, bounce (restitution, ball on board) 0.4, friction (ball on board) 0.3. "Mean" is the mean bin (6 for fair coins), "spread" the standard deviation (1.73 for fair coins). With 150 balls one run's mean scatters by about 0.15 to 0.2 from run to run, its spread by about 0.1.

1. The pitch of the pins

A ball must pass between two pins of a row, but the triangular lattice also has straight diagonals, and if those are wider than a ball, a ball moving a little sideways runs down one touching nothing. The first board (pins 3 mm, 13 mm apart, its diagonals 8.3 mm) did exactly that: traced, the first balls went from the neck to the outermost bins in straight lines. With 4 mm pins the window is narrow: between neighbours S - 4 must be more than 6, along a diagonal S sin 60 - 4 less than 6 or not much more.

Dropping the balls one at a time, 4 mm above the first pin (a random 2 to 4 mm either side), 150 balls:

pitch	between pins	diagonal	mean	spread
11 mm (4 mm pins)	7.0 mm	5.5 mm	6.05	1.42
12 mm (4 mm pins)	8.0 mm	6.4 mm	6.21	1.73
13 mm (3 mm pins)	10.0 mm	8.3 mm	5.80	2.72

At 11 mm the balls zigzag (alternate bins are favoured: each choice undoes the last a little), at 13 mm they jump pins. 12 mm gives the binomial's own spread, and is the board's. (That is for single balls dropped gently. Fed by the wheels, the same lattice gives about 1.95: section 6.)

2. The side walls

The side walls of the pins run where the next pin out would be. As smooth 60-degree walls they were ramps: a ball reaching one slid down it, squeezing past the outermost pins through a gap 0.5 mm narrower than itself (Bullet's soft contacts let a fast ball through a small overlap). So the walls have **half pins** standing out of them where those pins would be: a ball at the edge meets a pin head on.

A pin a little more than a ball's radius from a wall **cradles** a ball: one came to rest on the outermost pin of the 11th row, against the bins' wall (the pin's surface 3.2 mm from the wall: too little to pass, enough to sit on). A ball can't rest on a half pin, which pushes it away from the wall. So a pin that would leave less than a ball's width and a millimetre beside a wall is left out, and the wall has a half pin there; the bins' straight walls have one at every row.

3. Getting the balls in: a neck jams

Galton boards empty a hopper through a neck. Between two plates 7 mm apart the balls are a two-dimensional pile, and such piles clog:

- **A neck three balls wide** (18 mm), the first pin just under it: it jammed after two balls, every time. A ball came to rest on top of the first pin, held there by the balls pressing on it, and plugged the neck.
- **A straight throat** under the funnel, so that a ball perched on the pin leaves room beside it: it jammed after three, an arch across the throat's top standing on the perched ball. A taller throat (18 mm): 27 balls, then a zigzag column in the throat, standing on a perched ball that had rolled a little off centre, against a wall.
- **An obstacle above the neck** (as in Zuriguel and others' experiments with silos, 2011, it takes the weight off the arch): 19 balls before a jam instead of 2. Better, not cured.
- **Tapping**: the operator rocks the board a little whenever no ball has come through for 0.4 s. At 1.5 degrees it did nothing; at 4 degrees, alternating and growing while it stayed stuck, 102 balls and then a jam no tap cleared; and the taps threw the balls about and widened the spread.
- **A neck five balls wide** (30 mm, the lattice starting with three pins so that it fits): no clog at the neck, but the balls came so fast that they piled on the first row, whose 7 mm gaps arched over like a grate.
- **Wider gaps** (pins 13 mm apart): the balls flowed (150 in 2 s), but crowded together they knocked each other sideways and the bins came out nearly flat (a spread of 3.4).

So a crowd of balls spreads far wider than one ball does. On the same board, one ball at a time dropped low over the first pin: a spread of 1.78; from the bottom of a throat, as wide as the stream: 2.08; as a crowd from a wide neck: 2.68. The board needs its balls one at a time, and a neck can't do that without jamming.

A metering wheel can. Six slots in a wheel 32 mm across, under the hopper, turned by a crank: each slot picks up one ball from the bottom of the pile and carries it round under its housing (2 mm clear of the wheel: a slotted ball passes, a loose one can't get in). The hopper's downstream wall meets the wheel square on, so it scrapes off any ball half in a slot rather than pinching it. Five or six balls a turn. In every run here (well over fifty, with one wheel and with two) no wheel ever jammed, and the hand never had to turn the crank back.

4. The first pin wasn't a coin

The wheel hands every ball on in the same way, and that was the next trouble.

- **Dropped straight from the slot** onto the first pin: the slot is moving left at the bottom of its turn (9 cm/s), and the board leant left (mean 5.60).
- **Through a straight chute** (7.2 mm, a ball and 1.2 mm): each ball bounced once off the chute's left wall and left it moving right at 3 cm/s, and nearly every one went right at the first pin (means 6.53 and 6.49). With the chute 1.5 mm to the left, every one counted went left (5.27). A wider chute: all left (4.89). (With the chute 3 mm to the left, and with one only 0.6 mm wider than a ball, the runs stopped with a ball stuck somewhere; not traced.) The first pin was not a coin but a switch, set by the chute.
- **A funnel** (18 mm at its mouth, 8 mm at its neck): the balls bounce between its walls a few times, and the bounces make the most of the small differences between one ball and the next (where it sat in its slot). The first pin sent 47% of the balls counted to the left; over six runs the mean was 6.08 and the spread 1.95. But the balls still left it moving left, 5.4 cm/s on average, and how that came out at the first pin depended on the friction: at 0.15, 80% went left (means 5.19 to 5.65); at 0.45, two thirds went right (6.22 to 6.35).
- **A funnel and a straight neck** under it (8 mm wide, 8 mm long): the balls left the neck at 0 plus or minus 1 cm/s on average, but still 0.6 mm left of centre. Over six runs the means were 6.09 to 6.45 (6.31 on average): a lean of a third of a bin to the right; at a bounce of 0.3 it leant left instead (5.33 to 5.69). I first put that down to the bounce. It was the wheel. **Turned the other way,** every lean changed sides: at a bounce of 0.4, means 5.97 and 5.61; at 0.3, 6.63 and 6.66; and the balls left the

neck 0.6 mm right of centre (1.1 mm at 0.3). The pins and bins are symmetric (one ball at a time from rest, section 1, they gave the binomial's spread); the feed was not: the slot hands each ball on moving sideways, and spinning, and how that comes out at the first pin depends on the bounce.

- **Releasing the ball where the slot moves straight down** (3 o'clock) doesn't help: tried with the wheel alone, the ball rides the slot's lower side, which tilts as the wheel turns, and rolls out about 40 degrees below level, moving sideways at 7 cm/s and spinning at 41 radians a second, every ball exactly alike.
- **Two wheels, mirror images,** geared together so that they turn opposite ways on the one crank, each dropping its balls into one funnel over the neck, the right wheel's slots half a slot behind the left's: the machine is symmetric, so whatever one wheel's balls do, the other's do mirrored. The balls left the neck at 0.0 plus or minus 0.1 mm from centre. This is the board's; section 5 has what it gives.
- **Two balls over the neck jam.** At one turn a second the pair drops a ball every twelfth of a second, from opposite sides, and two can meet over the neck: once 16 balls piled up and arched over it, the funnel a small hopper again. At half a turn a second (a ball every sixth of a second) it has not happened; the crank goes up to 0.75.
- **The drawer was too small.** Pulling the slide, a few balls stayed at the slide's level, and knocking the board (below) did nothing: they were sitting on the heap in a full drawer. 40 mm deep holds 150 balls only if they lie flat, and they pour in heaped up in the middle; one wheel had got through by the luck of the packing. The drawer is 46 mm deep now.
- **The knock.** If balls stay wedged in a bin when the slide is out (a zigzag stack can brace itself against a bin's walls), the operator knocks the board (idealised: the balls in the bins get a jolt of 15 cm/s upward). Since the deeper drawer it has not been needed.

The tour checks each run: every ball in a bin, the mean within 0.6 of 6, the spread between 1.4 and 2.6.

5. Bounce, friction and speed

The finished board (the pair), each setting changed alone, three runs each (six at the usual settings: bounce 0.4, friction 0.3, the crank at half a turn a second); beside it, the one wheel and neck before (crank at one turn a second):

setting	means, the pair	spreads, the pair	tour	means, one wheel
usual	5.70, 5.81, 5.99, 6.06, 6.11, 6.17 (5.97)	1.77 to 2.04	pass	6.09 to 6.45 (6.31)
bounce 0.3	5.89, 5.97, 6.11	1.93 to 2.11	pass	5.33 to 5.69 (fail)
bounce 0.55	5.64, 5.91, 6.30	2.35 to 2.56	pass	6.07 to 6.51
bounce 0.7	6.13, 6.35, 6.41	2.46 to 2.76	fail (the spread)	6.05 to 6.42
friction 0.15	5.99, 6.03, 6.13	2.01 to 2.13	pass	5.51 to 6.10
friction 0.45	5.81, 5.82, 5.83	1.86 to 1.93	pass	6.09 to 6.18
crank 0.75	5.78, 5.94, 6.19	1.87 to 2.11	pass	

With one wheel the bounce chose the lean; with the pair the mean stays within about 0.3 of 6 from a bounce of 0.3 to 0.55, at either friction, and at three quarters of a turn a second. At 0.7 the means come out a little high (6.30 on average; one run's mean scatters by about 0.2 at that spread, so it may be chance or a small lean) and the spread reaches 2.76, past the tour's 2.6. The bounce sets the spread: about 1.9 at 0.3 and 0.4, about 2.5 above 0.5, where the balls jump pins. At one turn a second (a ball every twelfth of a second) the spread was wider too, up to 2.3: the balls crowd. In every run that finished, every ball reached a bin, no wheel jammed and the crank was never turned back.

The bounce is the number least known. No published coefficient of restitution for a steel ball on printed PLA at these speeds (a few tenths of a metre a second) was found; 0.4 is a guess. It is easy to measure: drop a ball from 100 mm onto a printed tile lying on a hard table, and see how high it comes back (a phone filming against a ruler); the bounce is the square root of that height over 100 mm. With the pair, the bounce no longer decides which way the board leans; it decides how wide the bins spread. Section 6 says why.

6. The pins are fair, but not independent

Six runs at the usual settings, pooled (900 balls): mean 5.97, spread 1.95 (1.73 for fair coins; its own uncertainty about 0.05). Against the binomial, pooling the outer three bins at each end, chi-square is 44.6 on 8 degrees of freedom, where 20 would already be one in a hundred: not chance. The tails hold 56 balls against 34.7, and the even bins are favoured (the middle bin 222 against 203, bins 5 and 7 below their share).

Recording each ball's path (`GB_PATHS`, with `paths.py`): at the level halfway below each row, which gap it passed. Four runs, 600 balls, 7,200 decisions, the usual settings:

	measured	fair coins
P(right), row by row	0.48 to 0.54	0.5
P(the same way as at the last pin)	0.661 (+-0.006)	0.5
the same, first pair alone	0.98	0.5
the same, from the third row on	0.63	0.5
correlation, decisions 1, 2, 3 rows apart	+0.34, -0.21, -0.10	0
jumped pins	0.6%	0

Each pin is fair; the pins are not independent. A ball leaves a pin still moving sideways and meets the next one off-centre on the same side. At the top it is nearly certain: a ball comes out of the neck straight down, the first pin throws it hard to one side and the second sends it on the same way. Two rows apart the sign turns: after twice the same way a ball is a little likelier to turn back, which partly cancels the persistence. The spread of the paths, built up from its parts (the variances add):

	spread
each decision alone (fair, but with the jumps)	1.77
+ the first pair going together	1.91
+ the neighbouring pins further down	2.28
+ pins two and three rows apart (negative)	2.04

So of the excess over fair coins, about a seventh is the jumps, about four tenths the first pair, and the rest the memory further down, net of its turning back. (The paths' bins are the logged bins, ball for ball; these four runs came out a little wider than the six above.)

What takes the memory away? Each setting four runs, 600 balls (450 where marked), the same instrument; V is the rows' spacing, 10.39 mm (S sin 60) on the board:

setting	P(same)	corr. 1, 2 rows	jumps	spread
bounce 0.1 (450)	0.91	+0.83, +0.63	0.1%	4.21
bounce 0.2	0.81	+0.63, +0.22	0.1%	2.87
bounce 0.3	0.70	+0.41, -0.15	0.0%	2.14

setting	P(same)	corr. 1, 2 rows	jumps	spread
usual (bounce 0.4)	0.66	+0.34, -0.21	0.6%	2.04
bounce 0.55	0.68	+0.41, +0.01	1.8%	2.55
friction 0.15	0.66	+0.33, -0.15	0.5%	2.10
friction 0.45	0.64	+0.30, -0.22	0.8%	2.01
friction 0.7	0.63	+0.29, -0.23	0.8%	1.86
bounce 0.1, friction 0.7	0.84	+0.69, +0.34	0.0%	3.05
V 8.3 (rows flatter)				balls perch in the pins; no run finished
V 9.35	0.84	+0.69, +0.37	0.5%	3.24
V 11.2	0.56	+0.19, -0.14	3.1%	2.13
V 11.95	0.51	+0.15, -0.05	6.3%	2.64
V 11.95, pins 5 mm	0.42	0.00, -0.03	9.3%	2.60
V 11.95, bounce 0.25	0.68	+0.37, -0.23	0.1%	1.96
V 11.2, bounce 0.25 (450)	0.70	+0.41, -0.17	0.2%	2.06
V 11.95, bounce 0.15 (450)	0.83	+0.66, +0.29	0.3%	3.25

- **A deader ball remembers more, not less.** Damping the pins (silicone sleeves, say) would make it worse. A ball that barely bounces rolls round a pin, leaves it moving sideways, and lands on the next pin's same side: at 0.1 it runs down a diagonal nine times in ten. K. Judd found the same in a model of Galton's quincunx ("Galton's quincunx: random walk or chaos?", 2006): at some restitutions every shot settles into moving steadily left or right, and even where the fraction looks fair the transitions are correlated.
- **Taller rows forget, but jump.** With more fall between rows the ball meets the next pin with less of its sideways speed, and the memory nearly goes (at 11.95 mm with 5 mm pins the correlation is nil). But a bounce now has room to carry a ball past the next pin: one decision in eleven a jump, and the spread is as wide as ever. Flatter rows are worse both ways, and at 8.3 mm balls perch.
- **Every way of losing the memory buys jumps.** A deader ball in taller rows comes back to where the board is (1.96). Only more friction helped a little on its own (1.86 at 0.7, a high figure for steel on PLA).

So the board as built, with its equilateral lattice (Galton's, apparently) and a bounce of 0.4, is already at about the narrowest spread these knobs give. The part that is not the lattice's is the first pair: a ball that reached the first pin moving a little sideways, rather than straight down out of the neck, might not repeat 98% of the time. That is left to try. The bounce of real balls on printed pins matters here more than anywhere (section 5 says how to measure it).

7. Settling, and resting

The balls low in a tall stack (36 balls in the middle bin) jitter at 5 or 6 cm/s at 20 solver passes: the solver's, not a movement (60 passes bring it down to 1). Rather than pay three times the passes, a ball counts as still under 8 cm/s; the run is over when the hopper is empty, nothing is on its way and everything has been still for half a second. The exhibit then rests (stops stepping) until the next key.

8. How fast it runs

While the balls are running, about 2.5 s of computing a simulated second here (headless, on one core); the tour, two runs of 28 s with the emptying and pouring between, about four minutes. At rest, nothing.

9. The printed parts

- Three tiles, each under 200 mm (166 x 171, 196 and 199 mm), joined by splices glued across the back.
- Every STL is one watertight solid, and the parts at rest don't overlap (all 45 pairs of the board, both wheels, both gears, the slide, drawer, front sheet, crank and stand: checked with manifold3d). The two gears mesh at their pitch circles, the right one turned half a tooth.
- Found on the way: where an outline only touched a circle that is cut as well (the wheel's recess), OpenSCAD left slivers, the extruded walls "were not closed", and a whole tile's walls went missing, in an STL that was still watertight (only the back plate was left). Each tile's thickness is checked now, and the outlines for crossing themselves. The drawer's bay and the slide's slot were first cut only halfway across: the overlap check found it.